

Dr. Babasaheb Ambedkar Technological University Lonere

SUMMER SEMESTER EXAMINATION - May 2018

IV Semester B. Tech. Civil Engineering

CV 401 Hydraulic Engineering II

Time: 03 Hours

Max. Marks: 60

Instructions to the Students

1. Attempt any five questions from Q. 1. To Q. 6.
2. Illustrate your answers with neat sketches, diagrams etc. where ever necessary. Data required for solving the question, but not mentioned therein, is part of examination. Therefore, Students shall make suitable assumptions.
3. Figure in parenthesis, on the right hand side of a question indicates marks of that question.

Q. 1 Solve :

- a) Show that the discharge formula for a trapezoidal channel having Manning's coefficient $N = 0.0126$ and carrying maximum flow is given by

$$Q = 100 y^{8/3} \left(\sqrt{1 + z^2} - \frac{z}{2} \right) S_b^{1/2}$$

Where Q is discharge in m^3/s ; y is depth of flow in meter in the channel of side slopes 1 vertical to z horizontal and S_b is the bed slope of the channel. (5)

- b) What size of circular drainage pipe is needed to carry $1.10 m^3/s$ of discharge when flowing half full? The pipe is laid at a slope of 1 in 2500 and Manning's N for the pipe is 0.018. (4)
- c) Calculate the specific energy E in a critical flow at depth y_c occurring in a triangular channel. (Calculate 'E' in terms of critical depth ' y_c ') (3)

Q. 2 Solve :

- a) Water flows in a triangular channel of side slope 1 horizontal to 1 vertical and longitudinal slope of 0.001. Determine the type of channel bottom slope (whether the channel has mild, steep or critical slope etc.) The discharge flows through it is $0.2 m^3/s$. Assume Manning's N is 0.015. For what range of depths will the flow be on a type 1, 2 and 3 curve? (Means determine the ranges of depths for zone 1, zone 2 and zone 3 for the given GVF profile) (5)
- b) A hydraulic jump is formed in a 2m wide rectangular open channel which is horizontal and frictionless. The post jump depth and velocity are 0.8m and 1m/s respectively. The pre-jump velocity is? (3)
- c) Explain the term 'Specific Force'. Show that the 'Specific Forces' at two different sections in a short horizontal reach of prismatic channel (in an open channel flow) are equal. (4)

Q. 3 Solve :

- Calculate the friction drag force on a plate 0.15 m wide & 0.45 m long placed longitudinally in a stream of oil having a free stream velocity of 6 m/s. Specific gravity of oil is 0.925. Total drag coefficient $C_D = 0.0153$ (3)
- The velocity distribution in the boundary layer is given by : $\frac{u}{U} = \frac{y}{\delta}$ where u is the velocity at a distance y from the plate and $u = U$ at $y = \delta$, δ being boundary layer thickness. Find the value of energy thickness. (3)
- Explain the term hydraulic jump. Derive an expression for the depth of hydraulic jump in terms of the upstream Froude number. (6)

OR

A Pelton wheel has a mean bucket speed 10 m/s with a jet of water flowing at the rate of 700 litres/s under a head of 30 meters. The bucket deflects the jet through an angle of 160° . Calculate the power given by water to the runner and hydraulic efficiency of the turbine. Assume coefficient of velocity as 0.98. (6)

Q. 4 Solve :

- A stream has a width of 25 m and depth of flow 2.5 m. The mean velocity of the flow is 1.10 m/s. Find the height of a weir to be built in the stream to raise the water level by 1.2 m. Assume coefficient of discharge as 0.90. (4)
- A right angled triangular notch is provided in the vertical side of a tank having plan area of 0.93 sq.m. uniform at all levels. When the head over the notch is 75 mm, it is found that the water surface in the tank is falling down at a rate of 2.54 mm per second. Calculate the coefficient of discharge of the notch. <https://www.batuonline.com> (4)
- If 'H' is the height of water above crest level and 'h' is the head of head of water at the middle of the weir, Derive the relation between H and h for the condition of maximum discharge over a broad crested weir. (4)

Q. 5 Solve :

- With the help of neat diagram explain the construction, components and working of a Francis turbine. (4)
- A turbine develops 8000 kW power under a head of 30 m when revolving at a speed of 100 rpm. Find its specific speed. If the head on the same turbine falls during the summer season to 18 m, find the power and speed developed by the turbine. (4)
- A pump can deliver discharge of $0.10 \text{ m}^3/\text{s}$ to a head of 30 m. The critical cavitation number for the pump is found to be 0.12. The pump is to be installed at a location where the barometric pressure is 96 kPa (abs) and the vapour pressure 3 kPa (abs). Assuming an intake pipe friction head of 0.3 m. Determine the minimum value of Net Positive Suction (4)

Head (NPSH). What would be the maximum allowable elevation above the sump water surface at which the pump can be located? (4)

Q. 6 Solve :

- a) Derive the following expression for force F exerted by a jet of area ' A ' which strikes a flat plate at an angle θ to the normal to the plate with velocity V . The plate itself is moving with velocity u in the direction of normal to the plate surface.

$$F = \rho A \frac{(V \cos \theta - u)^2}{\cos \theta} \quad (4)$$

- b) A jet of water of diameter 40 mm moving with a velocity of 30 m/s, strikes a curved fixed symmetrical plate at the centre. Find the force exerted by the jet water in the direction of the jet if the jet is deflected through an angle of 120° at the outlet of the curved plate. (4)
- c) A square plate of height 200 mm is suspended vertically from one of its edges using a hinge support as shown in figure. A water jet of 20 mm diameter having a velocity of 10 m/s strikes the plate at its midpoint, at an angle 30° with the vertical. Consider $g = 9.81 \text{ m/s}^2$ and neglect the self weight of the plate. Calculate the force required at the bottom of the plate to keep it in its vertical position? (4)

